

Dataset Description

1. Source and study sites

The data used in this study are hourly meteorological and solar-radiation records obtained from the NASA Prediction Of Worldwide Energy Resources (POWER) project, which derives surface solar-radiation fields from satellite observations and assimilated meteorological reanalysis. Records were retrieved for five provinces of Türkiye — Ankara, Antalya, Konya, Rize and Van — selected to span distinct climatic and topographic settings rather than to represent a single region. Their coordinates and elevations are given in Table 1.

Table 1. Study sites.

Province	Latitude (°N)	Longitude (°E)	Elevation (m)	Setting
Ankara	39.93	32.86	890	Central Anatolian plateau, semi-arid
Antalya	36.90	30.69	30	Mediterranean coast
Konya	37.87	32.48	1020	Central Anatolian steppe, continental semi-arid
Rize	41.02	40.52	5	Eastern Black Sea coast, humid subtropical
Van	38.49	43.38	1725	Eastern Anatolian highland, continental

The site selection is deliberately asymmetric, and the analysis below shows that the asymmetry is larger than the geographic spread suggests: four of the five sites form a single radiative regime, while Rize constitutes a second one. This is treated as a property of the dataset rather than as a limitation, because it is precisely the configuration under which a model's ability to transfer across climates can be tested.

2. Temporal coverage and sampling

Each site is represented by an uninterrupted hourly series spanning 30 June 2019 00:00 to 30 May 2026 23:00, comprising 60,648 hourly observations per site — 2,527 days, or approximately 6.92 years — and 303,240 observations in total. Coverage is identical across sites: the five series share the same timestamps, with no gaps, no duplicate timestamps and no missing values after the preprocessing described in Section 4.

The record terminates on 30 May 2026 because the provider's near-real-time processing latency leaves the final 744 hours of the retrieved file without validated irradiance values. Those hours were removed; the meteorological variables are complete to the end of the retrieved file, so the truncation is governed by the target variable alone.

2.1 Time convention

An aspect of the data that materially affects any diurnal analysis is that the hourly timestamps are **not expressed in a single time zone**. Each site is recorded in its own whole-hour local solar time, corresponding to UTC offset by the nearest hour to its longitude.

This was verified from the data itself. Taking the irradiance-weighted centre of mass of the mean diurnal profile as an empirical solar-noon estimate yields

$$| \text{Konya } 11.24 \approx \text{Ankara } 11.24 < \text{Antalya } 11.41 < \text{Van } 11.58 < \text{Rize } 11.91$$

which is the **reverse** of the ordering that a shared national clock would produce — under a common clock the easternmost sites would peak earliest — and reproduces the longitude-derived expectation to within 0.11 h at every site.

Two consequences follow and are observed throughout the analysis. First, a given hour index does not denote the same physical instant at different sites, so hourly quantities are never pooled or compared across sites; each site is presented in its own panel. Second, the cyclical encoding of the hour of day used as a model input is more informative under this convention than it would be under a shared clock, because each site is implicitly encoded in its own solar time. Hour labels denote the start of the interval they cover; all diurnal figures are plotted at interval centres and axes are labelled in local solar time.

3. Variables

The retrieved dataset provides one radiation variable, which is used as the forecast target, and six meteorological variables used as predictors, together with the calendar fields required to construct the timestamp. Table 2 reports pooled descriptive statistics over the complete record.

Table 2. Descriptive statistics of the retrieved variables, pooled over all five sites and all 303,240 hourly observations.

Variable	Description	Unit	Mean	SD	Min	Max	Skewness
ALLSKY_SFC_SW_DWN	All-sky downward shortwave irradiance at the surface (target)	W/m ²	193.75	275.23	0.00	1216.67	1.28
T2M	Air temperature at 2 m	°C	11.96	10.38	−24.33	42.30	0.06
RH2M	Relative humidity at 2 m	%	64.84	24.45	3.36	100.00	−0.37
T2MDEW	Dew-point temperature at 2 m	°C	3.93	7.37	−30.33	23.81	−0.19
PS	Surface pressure	kPa	88.32	6.04	75.74	97.73	−0.65
WS2M	Wind speed at 2 m	m/s	2.15	1.42	0.00	13.65	1.39

Variable	Description	Unit	Mean	SD	Min	Max	Skewness
WD2M	Wind direction at 2 m	°	— (circular; see Section 8.2)				
PRECTOTCORR	Bias-corrected total precipitation	mm/hour	0.073	0.282	0.00	14.49	9.45

Two variables require comment before any statistic drawn from them is interpreted.

Surface pressure does not behave as a meteorological variable in a pooled setting. Its pooled standard deviation is 6.04 kPa while its between-site standard deviation is 6.72 kPa: essentially all of its variance lies between sites rather than within them (site means range from 77.7 kPa at Van to 96.0 kPa at Antalya), reflecting elevation rather than weather. Its within-site variation, with a standard deviation of roughly 0.4–0.5 kPa, is the genuine synoptic signal. In a pooled model, surface pressure therefore functions largely as an elevation indicator, and this is the reason its apparent relationship with irradiance reverses once site and season are held fixed (Section 9).

Precipitation is effectively a binary variable at hourly resolution. Exactly 66.7% of daylight hours record zero precipitation, and the non-zero tail is strongly right-skewed (skewness +9.4 over all hours). A binary wet/dry indicator is more strongly associated with irradiance than the recorded amount at three of the five sites, while at Rize a logarithmic transform of the amount is the stronger of the two (Table 6).

Dew-point temperature is not an independent measurement. It can be reconstructed from air temperature and relative humidity through the Magnus relation with a Pearson correlation of 0.99919 and a root-mean-square deviation of 0.30 °C over all hours (0.99957 and 0.23 °C over daylight hours) — that is, at the level of the data's own rounding. It is retained for completeness but carries no information beyond the two variables from which it is derived. This redundancy is invisible to pairwise correlation analysis, because it is a two-variable relationship: the pairwise correlation between air temperature and dew point is only 0.61 (Section 9.2).

4. Data quality and preprocessing

Missing values. The provider encodes missing observations with a sentinel value. After removal of the 744-hour near-real-time tail described in Section 2, no sentinel value and no missing entry remains in any variable at any site. No imputation was performed, and no observation was interpolated.

Units. The retrieved irradiance is expressed in MJ/m² per hour and was converted to W/m² for consistency with the solar-forecasting literature; the conversion is exact and linear. Precipitation is retained in mm/hour, the natural unit for an hourly record. Summing the precipitation column over complete calendar years yields site totals of 340 mm (Ankara), 326 mm (Konya), 343 mm (Van), 664 mm (Antalya) and 1399 mm (Rize) per year, which reproduce the correct ordering and order of magnitude of Turkish climate

normals and confirm the unit interpretation. Consistent with the known behaviour of satellite-derived precipitation products, the values under-estimate observed normals at the two wettest sites.

Numerical resolution. The retrieved irradiance is stored to two decimal places in its source unit, which after conversion places the target on a discrete grid with a step of 2.78 W/m^2 ; the entire record contains 387 distinct target values. The resulting quantisation noise has a standard deviation of 0.80 W/m^2 , below 1% of the error of the weakest reference forecast considered here, and is therefore immaterial to the reported accuracy metrics. It is stated because it sets a floor on the resolution at which any irradiance difference in this dataset can be interpreted, and because it affects how daylight is best identified (Section 5).

Physically inadmissible values. One observation in the record (Van, 17 February 2020, 15:00 local solar time) reports 1216.67 W/m^2 , which exceeds the extraterrestrial irradiance incident on a horizontal surface at that site and hour (544 W/m^2) by more than a factor of two. It is physically impossible and is treated as a retrieval or back-fill artefact. It was neither removed nor modified, since a single observation in 303,240 cannot affect any aggregate reported here, but it must not be used to support any claim about the radiative extremes of the site.

5. Definition of daylight hours

Roughly half of all hourly observations are night hours in which the target is exactly zero. These observations are trivially predictable and, if included, dominate aggregate error statistics (Section 12). Separating them therefore requires an unambiguous criterion, and the criterion used here is **geometric**: an hour is classified as daylight if the apparent elevation of the Sun at the midpoint of that hour, computed from the site coordinates and the timestamp using a standard solar-position algorithm, is positive. Under this definition 152,893 of 303,240 observations (50.42%) are daylight, with a mean daylight duration of 12.07–12.14 h per day across sites.

Two alternative criteria were examined and rejected.

The first is a threshold on the observed irradiance itself. On this dataset it agrees with the geometric criterion on 303,204 of 303,240 observations, yet it is inadmissible on two grounds. It defines the evaluation subset using the quantity being predicted, so that a heavily overcast twilight hour recording zero is silently excluded — that is, observations are removed preferentially from the conditions under which a forecast is hardest. More decisively, it cannot be evaluated prospectively: a day-ahead forecast must determine whether the Sun will be above the horizon at a future hour without access to the observation at that hour, so a criterion based on the observation cannot be applied operationally. A geometric criterion is therefore required in any case, and adopting a different one for evaluation would be internally inconsistent.

The second is a climatological criterion based on the mean irradiance of a (site, month, hour) cell. This is too coarse: sunrise and sunset shift by 30–60 minutes within a calendar month, so the boundary hour of a cell is illuminated for part of the month and dark for the

remainder, and a cell mean classifies the entire hour as daylight. Applied to this dataset it admitted 5,266 observations occurring when the Sun was below the horizon.

The geometric threshold was not calibrated against the observations. Sweeping it produces a value that agrees more closely with the recorded irradiance, but adjusting a geometric criterion to fit the observed target reintroduces exactly the dependence the criterion is intended to avoid. The cost of leaving it uncalibrated was quantified and is small: the error of the strongest reference forecast changes by approximately 1%.

The residual disagreement between the geometric criterion and the recorded values is small and falls in the conservative direction. One daylight-classified observation records exactly zero. Conversely, 2,968 observations classified as night carry non-zero irradiance; these are twilight hours in which sunrise occurs part-way through the interval, and together they account for 0.024% of total daylight energy. Excluding them makes the daylight subset marginally harder to predict, not easier.

6. Statistical characterisation of the target

6.1 Distribution

Over daylight hours and pooled across sites, irradiance has a mean of 384.2 W/m², a median of 341.7 W/m² and a standard deviation of 277.7 W/m², with skewness +0.42 and excess kurtosis −0.94. Over all 24 hours the mean falls to 193.7 W/m² and the median to 8.3 W/m², with skewness +1.28.

The contrast between these two summaries reflects the mixture structure of the unconditional distribution: a point mass of exact zeros at night superimposed on the daylight distribution. The daylight distribution itself is **platykurtic** — broad and flat rather than peaked — which is the direct consequence of solar geometry sweeping irradiance from zero to approximately 1000 W/m² within each day. Figure 1 shows the per-site daylight distributions.

Distribution of daylight hourly irradiance

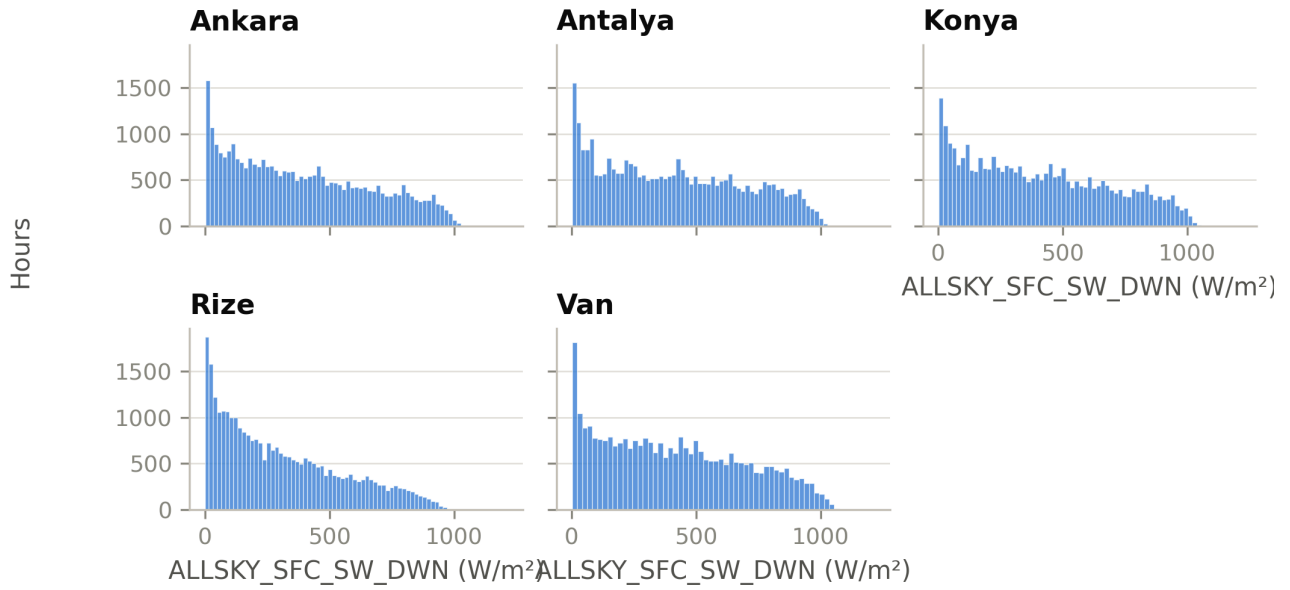


Figure 1. Distribution of hourly irradiance over daylight hours, by site.

Site-level statistics are given in Table 3.

Table 3. Target variable by site, daylight hours.

Site	N	Mean (W/m ²)	SD (W/m ²)	Median (W/m ²)	Max (W/m ²)	Skewness	Daily total (kWh/m ² /day)
Ankara	30,540	386.93	277.06	341.67	1027.78	0.42	4.68
Antalya	30,555	410.97	283.53	388.89	1044.44	0.28	4.97
Konya	30,502	405.25	282.42	369.44	1055.56	0.36	4.89
Rize	30,628	306.26	246.20	250.00	988.89	0.69	3.71
Van	30,668	411.60	282.89	383.33	1216.67	0.33	5.00
Pooled	152,893	384.18	277.67	341.67	1216.67	0.42	4.65

6.2 Diurnal and seasonal structure

Hour of day alone accounts for 73.1% of the variance of the target over all 24 hours, and for 48.8% within the daylight subset; day of year accounts for 8.6% and 14.8% respectively. A harmonic (sine-cosine) representation of each recovers essentially the whole of this explained variance (for example 0.7285 against 0.7313 for hour of day over 24 hours), which supports the use of cyclical rather than categorical encodings of time.

The practical implication is that roughly three quarters of an aggregate score computed over all 24 hours reflects knowledge of the day-night cycle rather than any forecast skill, and that within daylight hours the seasonal contribution approximately doubles in relative importance.

Seasonal means and variability are summarised in Table 4, and the diurnal profile by season is shown in Figure 2.

Mean diurnal irradiance profile by season (band: between-day IQR)

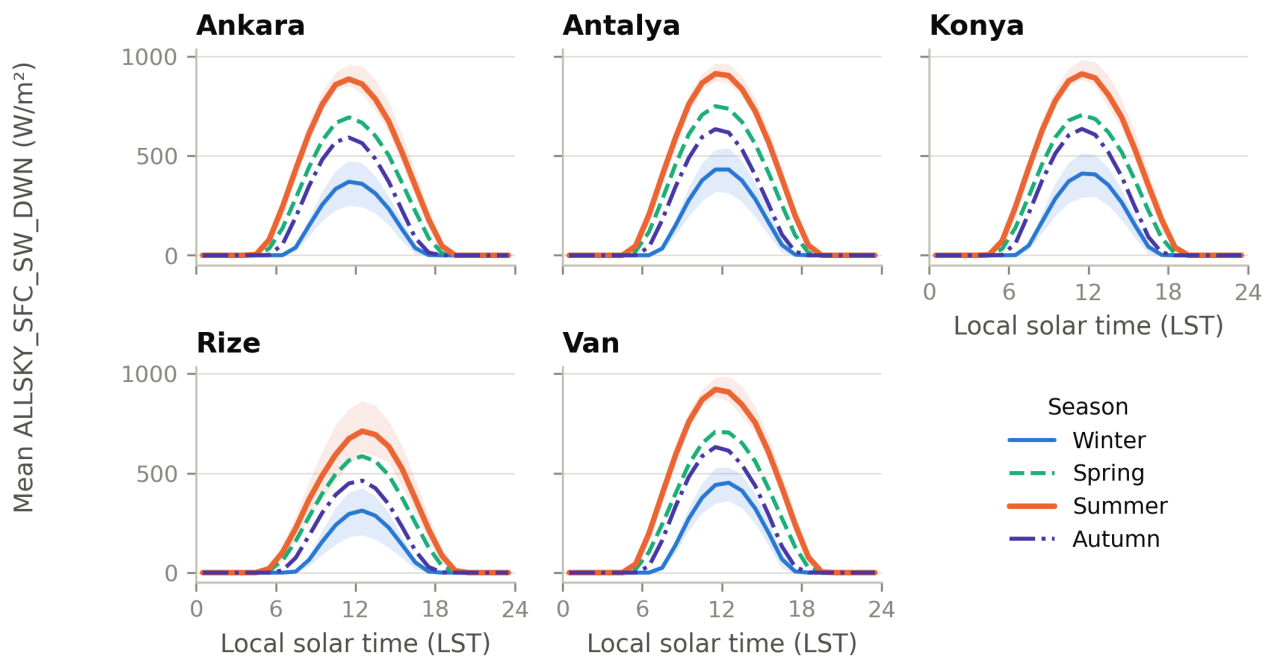


Figure 2. Mean diurnal irradiance profile by season and site, plotted in local solar time. The shaded band shows the between-day interquartile range for winter and summer.

Table 4. Seasonal characteristics, averaged over sites. Meteorological seasons are used (winter = December–February).

Season	Mean irradiance, 24 h (W/m ²)	Mean irradiance, daylight (W/m ²)	Daily total (kWh/m ² /day)	Between-day CV
Winter	98.4	235.4	2.36	0.409
Spring	218.5	400.4	5.24	0.340
Summer	296.1	501.2	7.11	0.154
Autumn	164.6	353.8	3.95	0.372

A structurally important feature emerges from the last column: **irradiance and its predictability move in opposite directions across the year**. Summer days are not only brighter but markedly less variable between days. Expressed per site, the ratio of winter to summer coefficient of variation is 3.04 at Ankara, 3.82 at Antalya, 3.05 at Konya, 2.72 at Van and 1.81 at Rize. Any error metric computed over the full year therefore aggregates two regimes of very different difficulty, and a small absolute error in winter does not indicate stronger performance in winter — there is simply less to predict.

6.3 Inter-annual variability

Over the six complete calendar years contained in the record (2020–2025), the mean daily total varies little from year to year: the relative standard deviation is 1.8–3.3% depending on the site, and the spread between the best and worst year ranges from 5.1%

(Konya) to 8.0% (Van). Rize is again the exception at 10.3%. Table 5 reports the annual means; Figure 3 presents the same information as a month-by-year anomaly field, which separates the inter-annual signal from the seasonal cycle that dominates the raw surface.

Monthly irradiance anomaly: departure from that month's 6-year mean

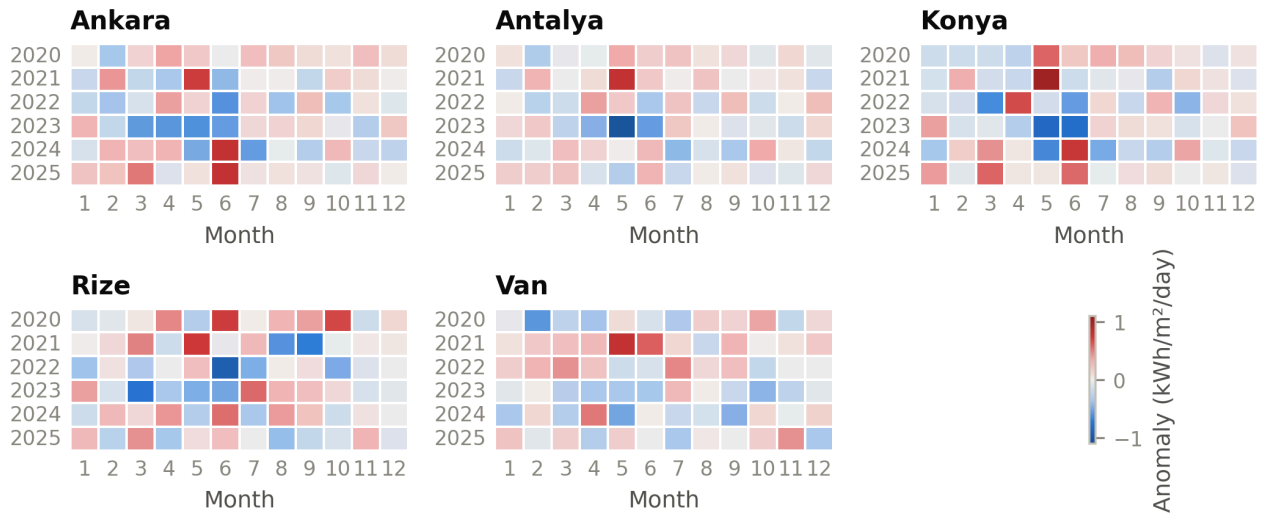


Figure 3. Month-by-year irradiance anomaly, expressed as the departure of each month from its six-year mean, by site.

Table 5. Mean daily total irradiance (kWh/m²/day) by complete calendar year.

Site	2020	2021	2022	2023	2024	2025	Best/worst spread
Ankara	4.83	4.73	4.64	4.56	4.72	4.90	7.2%
Antalya	5.06	5.13	5.04	4.86	4.99	5.04	5.6%
Konya	5.01	4.98	4.87	4.84	4.90	5.09	5.1%
Rize	3.91	3.75	3.54	3.68	3.82	3.74	10.3%
Van	4.95	5.26	5.16	4.87	4.95	5.05	8.0%

7. Clearness index and regional differentiation

Comparing sites on irradiance alone confounds latitude with atmospheric condition. The comparison is therefore made on the **clearness index**, defined in the standard way as the ratio of measured global horizontal irradiance to the extraterrestrial irradiance incident on a horizontal surface at the same instant,

$$kt = GHI / (I_0 \cos \theta_z),$$

where I_0 is the extraterrestrial normal irradiance including the Earth-Sun distance correction and θ_z is the solar zenith angle. The denominator is a purely astronomical quantity, so the index isolates atmospheric attenuation. Hours with an extraterrestrial irradiance below 20 W/m² are excluded, as the ratio is numerically unstable at twilight.

The index is well behaved on this dataset: its median is 0.555, its 99th percentile is 0.801, and only 2 of 150,746 illuminated hours exceed unity. No clipping or truncation was applied. Sky conditions are classified using the conventional bands for this index, namely clear above 0.65 and overcast below 0.35.

Table 6 shows that the five sites do not form a single population.

Table 6. Radiative regime by site. Clearness index values are daily.

Measure	Ankara	Antalya	Konya	Van	Rize
Daily total (kWh/m ² /day)	4.68	4.97	4.89	5.00	3.71
Mean daily clearness index	0.574	0.589	0.589	0.609	0.463
Median hourly clearness index	0.565	0.583	0.586	0.600	0.422
Clear-day share (kt > 0.65)	43.8%	43.7%	47.4%	50.0%	13.8%
Overcast-day share (kt < 0.35)	11.3%	8.8%	10.6%	6.4%	28.9%
Between-day coefficient of variation	0.489	0.436	0.457	0.445	0.566

Four of the sites differ from one another by 7% in mean daily total and occupy a narrow band on every measure in the table. Rize lies 21–26% below that band in level and, more importantly, in a different position on every measure of variability: it experiences overcast days between two and four times as often as any other site, and reaches clear-sky conditions on roughly one day in seven against one day in two. Its **best** season (summer, mean clearness 0.522) is comparable to the other sites' **winter** (0.477–0.556). The difference is therefore one of regime, not of degree. Figure 4 summarises this comparison.

Rize is a separate regime from the other four provinces

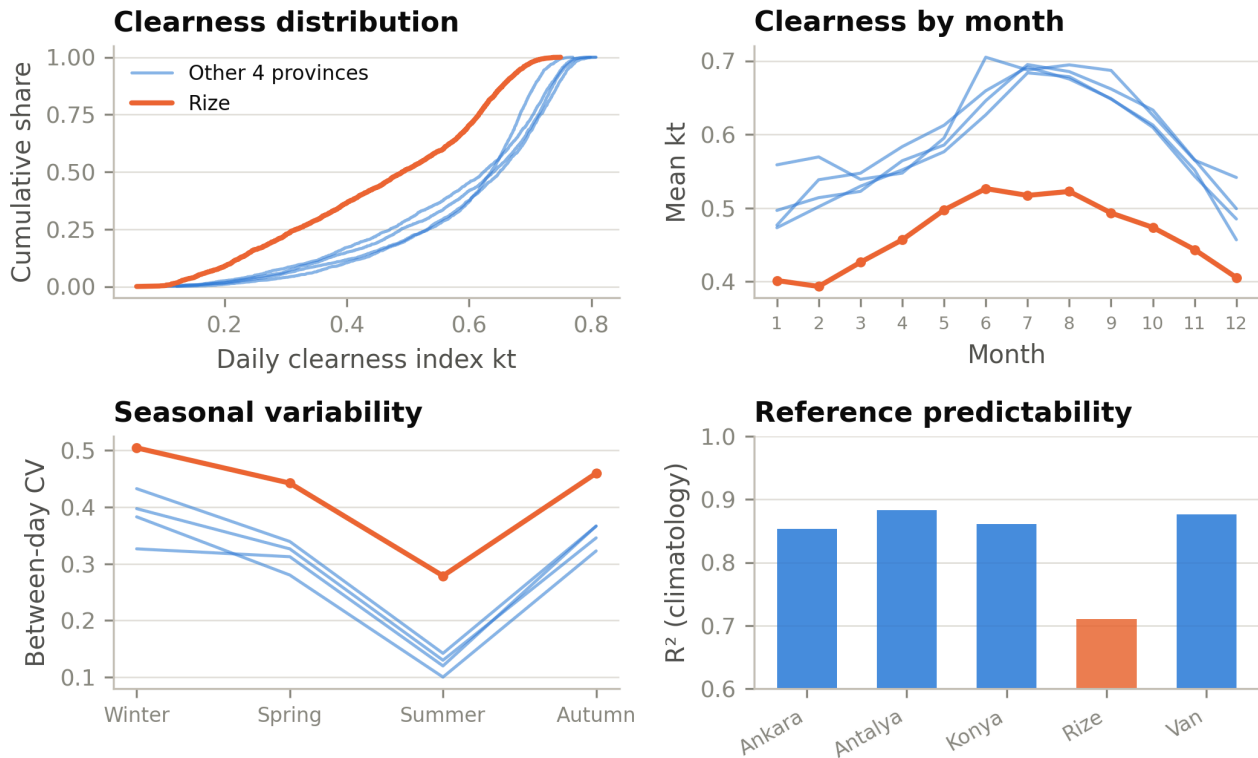


Figure 4. Radiative regime of Rize against the other four sites: cumulative distribution of the daily clearness index, the monthly clearness cycle, seasonal between-day variability, and the accuracy of the climatological reference forecast.

Van occupies the opposite extreme: the highest daily total, the highest clearness index, the lowest overcast-day share and the lowest winter variability of the five, consistent with its combination of high elevation and dry continental air.

8. Meteorological structure

8.1 Hour-to-hour variability

The distribution of absolute change between consecutive daylight hours is summarised in Table 7. Most of the variability in raw irradiance is geometric — the Sun rising and setting — so the informative column is the change in clearness index, which removes that component.

Table 7. Hour-to-hour variability over daylight hours.

Site	Median $ \Delta \text{ irradiance} $ (W/m ²)	90th pct	99th pct	Share > 200 W/m ²	99th pct $ \Delta \text{ clearness} $
Ankara	105.6	188.9	211.1	3.4%	0.222
Antalya	116.7	191.7	219.4	6.0%	0.218
Konya	111.1	194.4	216.7	6.1%	0.222
Rize	83.3	166.7	213.9	1.7%	0.206

Site	Median $ \Delta \text{ irradiance} $ (W/m ²)	90th pct	99th pct	Share > 200 W/m ²	99th pct $ \Delta \text{ clearness} $
Van	116.7	194.4	216.7	6.7%	0.224

On the clearness scale the five sites are strikingly similar (0.206–0.224 at the 99th percentile). Rize's smaller raw ramps therefore reflect a weaker radiative envelope rather than a steadier atmosphere. Figure 5 shows the cumulative distributions.

Cumulative distribution of hourly irradiance ramps (daylight)

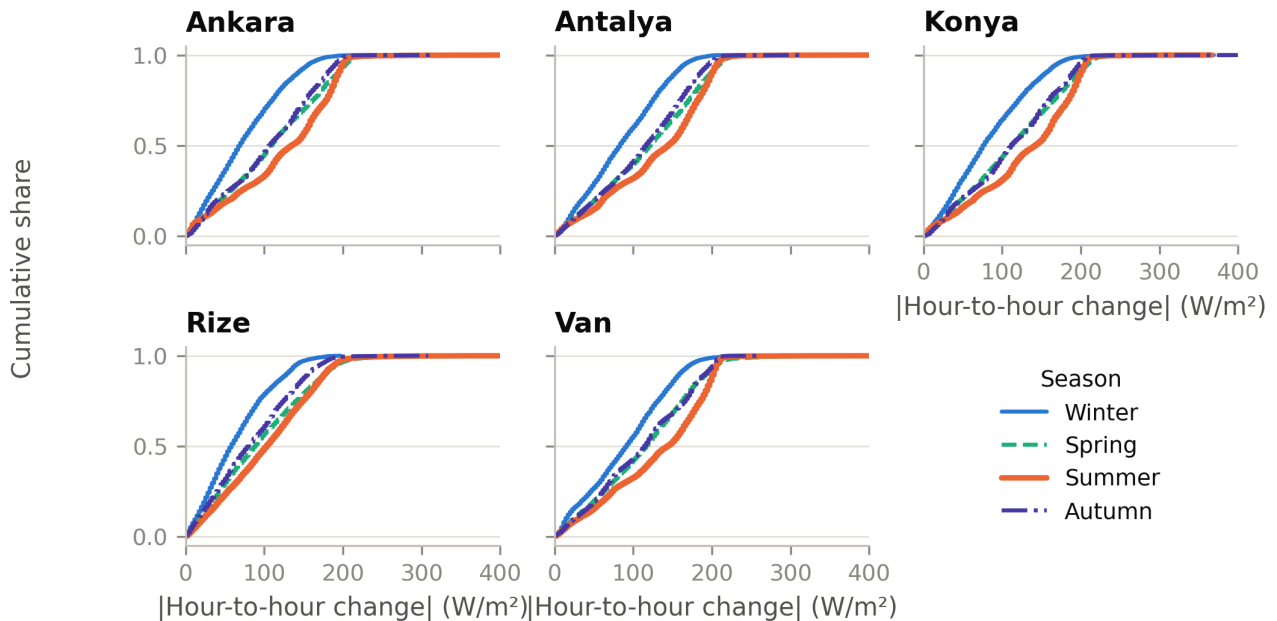


Figure 5. Cumulative distribution of the absolute hour-to-hour change in irradiance over daylight hours, by season and site.

8.2 Wind direction

Wind direction is a circular variable whose arithmetic mean is not meaningful, and it is summarised using speed-weighted circular statistics with calm hours (speed at or below 1 m/s) excluded. The resultant length R ranges from 0, for a completely dispersed distribution, to 1 for a single prevailing direction.

Table 8. Circular statistics of wind direction.

Site	Mean direction (°)	Resultant length R	Circular SD (°)
Van	215	0.468	71
Rize	270	0.244	96
Konya	337	0.227	99
Antalya	43	0.188	105
Ankara	332	0.124	117

Only Van exhibits a pronounced prevailing direction. At the remaining four sites the distribution is close to uniform, and wind direction carries correspondingly little information about irradiance. The share of calm hours differs substantially between sites (from 8,606 at Konya to 16,175 at Rize), and direction in those hours is essentially noise; this exclusion should be borne in mind whenever direction statistics are compared across sites.

9. Relationships among variables

9.1 Raw and partial association with the target

A correlation computed directly between a meteorological variable and irradiance is confounded with solar geometry, because most meteorological variables themselves follow the diurnal and seasonal cycles. To separate the two, the correlation was recomputed after removing the mean of each (site, month, hour) cell, which holds solar geometry and season fixed. Table 9 contrasts the two.

Table 9. Correlation with irradiance, pooled over daylight hours.

Variable	Raw r	Partial r (within site-month-hour)
RH2M	−0.628	−0.530
T2M	+0.515	+0.307
PRECTOTCORR	−0.168	−0.328
T2MDEW	+0.042	−0.272
PS	−0.038	+0.268
WS2M	+0.143	−0.150

Three variables reverse sign and the association of precipitation approximately doubles. The mechanism is straightforward: warm, windy, high-dew-point hours are predominantly summer midday hours, when irradiance is already high for geometric reasons. Removing that confound eliminates the spurious association and reveals the physical one, in which moisture and cloud reduce irradiance.

The strongest evidence that the partial formulation is the correct one is its behaviour across sites. Under raw correlation, surface pressure and wind speed take **inconsistent signs** at different sites. Under partial correlation, **all six variables take the same sign at all five sites**. Once solar geometry is removed, the sites agree about the physics. Figure 6 presents the correlation structure.

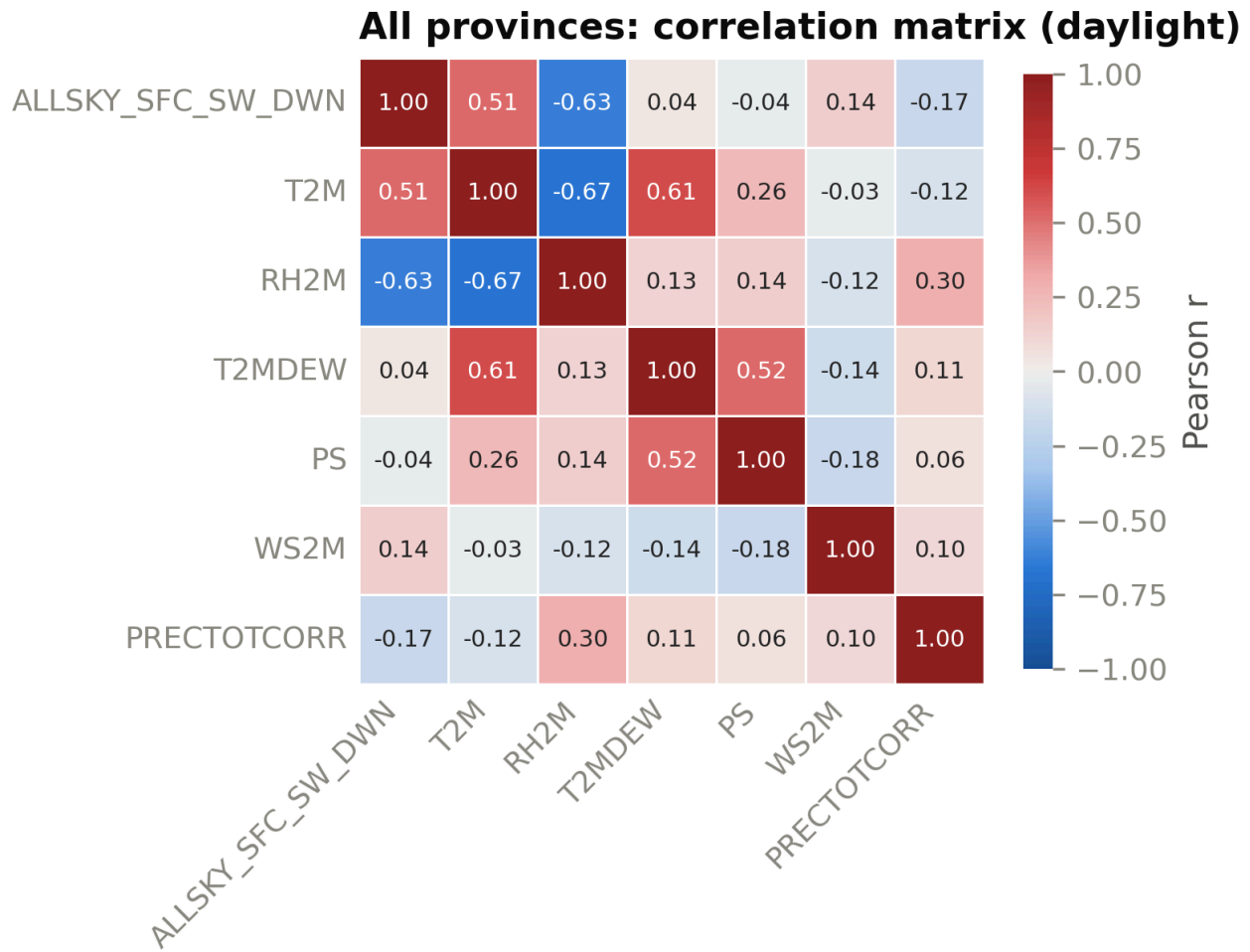


Figure 6. Correlation matrix of the target and the meteorological variables over daylight hours, pooled across sites.

Relative humidity is the only variable that is strongly associated with irradiance under both formulations, and it is also the variable with the largest between-site dispersion in Table 2. It is, on this dataset, the single most informative meteorological predictor.

9.2 Linearity and redundancy

Spearman and Pearson coefficients agree closely: the largest discrepancy against the target is -0.059 for precipitation, followed by $+0.047$ for wind speed, and no variable exceeds 0.06 . There is therefore no evidence of a non-monotonic relationship. That the discrepancy concentrates in precipitation and wind speed is expected, both being strongly skewed, and the fact that the rank correlation is the larger of the two supports treating precipitation on a rank or indicator scale rather than as a linear quantity.

Among the predictors, no pair exceeds a correlation of 0.9 . The pairs exceeding 0.5 are air temperature with relative humidity (-0.673), air temperature with dew point ($+0.609$) and dew point with surface pressure ($+0.521$), all of which are physically expected and none of which indicates redundancy at a level requiring removal.

This pairwise result should not, however, be read as evidence that the variable set is free of redundancy. As noted in Section 3, dew-point temperature is an exact function of air temperature and relative humidity, reproducible to $0.30\text{ }^{\circ}\text{C}$; pairwise correlation cannot

detect this because the relationship involves two variables jointly. A collinearity screen of this kind provides a lower bound on redundancy, never an upper one.

10. Temporal dependence

The autocorrelation structure was examined on the clearness index rather than on raw irradiance. The autocorrelation of raw irradiance is dominated almost entirely by the diurnal cycle and is therefore uninformative; dividing out the astronomical component leaves the persistence of atmospheric condition, which is the quantity a forecast must actually exploit. Figures 7a and 7b show both resolutions.

Hourly autocorrelation of the clearness index (dotted: 24 and 48 h)

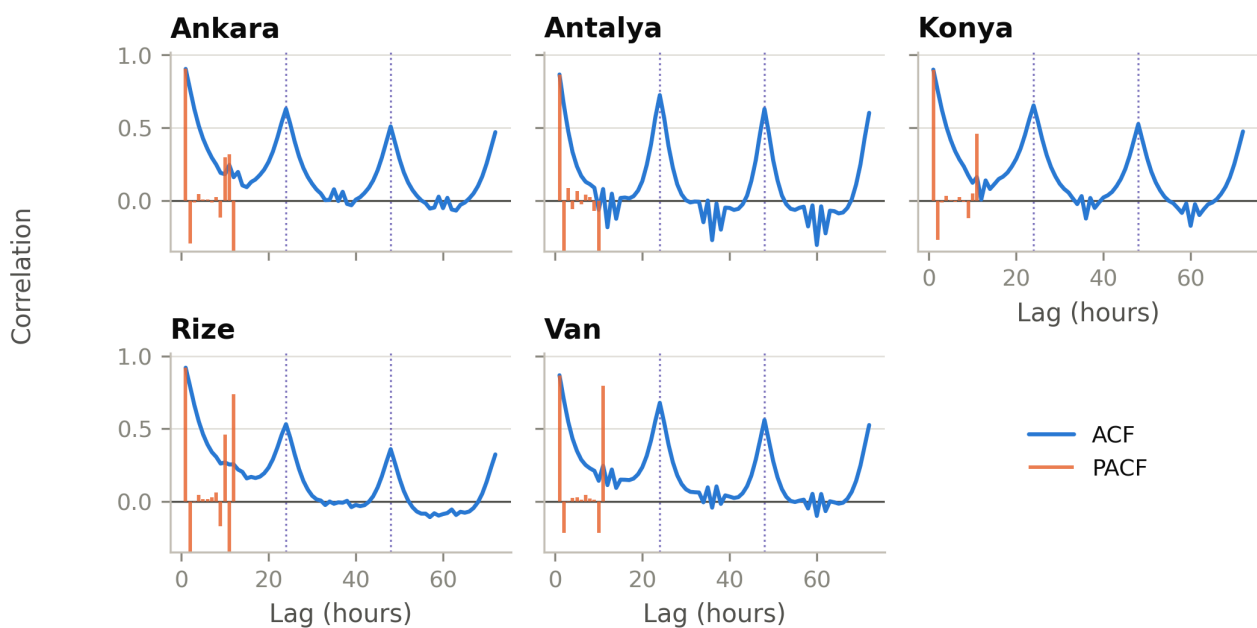


Figure 7a. Autocorrelation and partial autocorrelation of the clearness index at hourly resolution, by site. Dotted lines mark lags of 24 and 48 h.

Daily autocorrelation of the clearness index

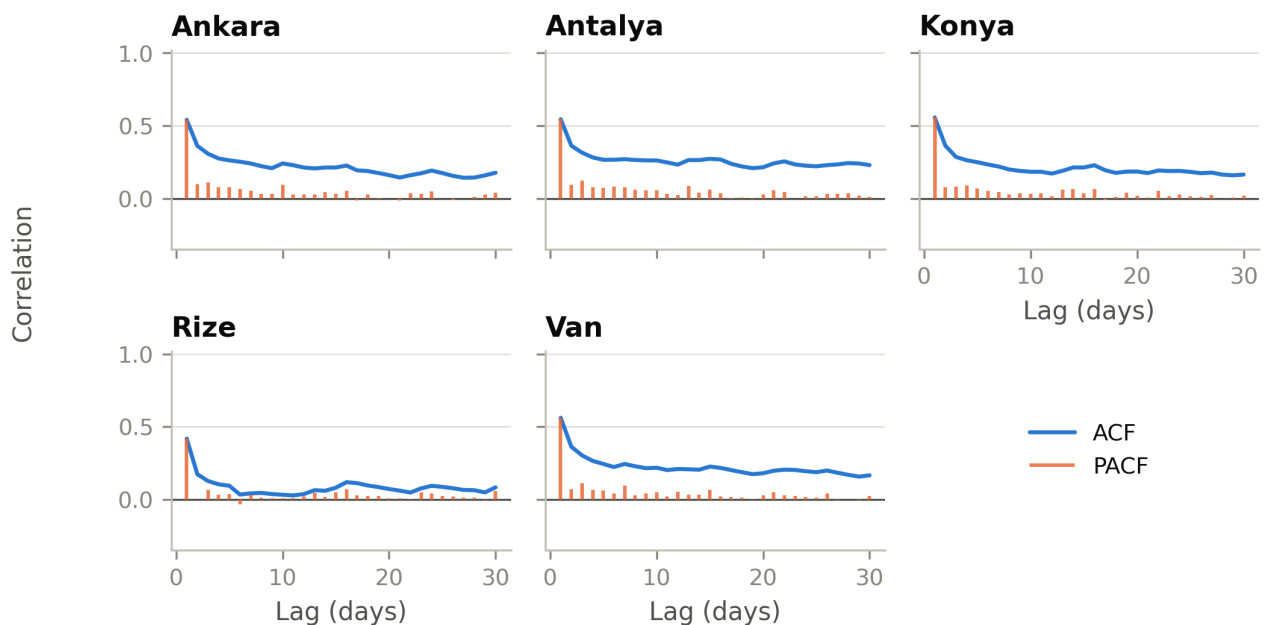


Figure 7b. Autocorrelation and partial autocorrelation of the clearness index at daily resolution, by site.

Table 10. Partial autocorrelation of the clearness index.

	Ankara	Antalya	Konya	Rize	Van
Hourly, lag 1	0.905	0.866	0.900	0.920	0.867
Hourly, lag 2	−0.295	−0.373	−0.270	−0.345	−0.215
Hourly, lag 3	−0.013	+0.087	−0.016	−0.006	−0.006
Daily, lag 1	0.541	0.545	0.557	0.417	0.561
Daily, lag 2	0.098	0.094	0.076	−0.002	0.069
Daily, lag 3	0.112	0.122	0.082	0.066	0.110

At the hourly resolution the series behaves as a second-order autoregressive process: the first-lag partial autocorrelation exceeds 0.87 at every site, the second lag is distinctly negative, and the third is indistinguishable from zero.

The daily resolution is the one that governs a day-ahead forecast, and there the decay is sharp. The partial autocorrelation falls from 0.417–0.561 at one day to between −0.002 and 0.098 at two days — a five- to six-fold reduction. Beyond the first lag, the information available to a day-ahead forecast from the target's own history is close to exhausted. The residual autocorrelation still visible at a lag of 30 days (0.08–0.23) reflects the seasonal cycle rather than memory, and is already captured by the cyclical encoding of day of year.

Rize again lies outside the band on every line, and consistently in the direction of shorter memory: its atmospheric condition is not only more variable but also less persistent.

An additional structural property constrains any multi-hour forecast on this dataset. Daylight hours occur in uninterrupted daily blocks of median length 12 h (minimum 9, maximum 15), and no block reaches 24 h at any site. A forecast horizon of one day therefore **always** spans at least one night period, so approximately half of every forecast window is determined by geometry alone.

11. Data partitioning

The record is divided chronologically, with the oldest observations used for training and the most recent for testing, so that no future information is available at training time. The boundaries fall on identical dates at all five sites, so every site is split on the same calendar and the partitions remain comparable across sites.

Table 11. Chronological partitions.

Partition	Period	Hours per site	Days
Training	30 Jun 2019 – 11 Aug 2024	44,879	1,870
Validation	12 Aug 2024 – 16 May 2025	6,671	278
Test	16 May 2025 – 30 May 2026	9,097	379

The proportions were chosen so that the test partition exceeds one full year, which it does at 379 days. This is a deliberate constraint rather than an arbitrary split: a test period shorter than a year would sample the seasonal cycle unevenly and bias the reported error towards whichever seasons it happened to contain. As partitioned, the test period covers 13 calendar months and all four seasons, with 12,725 spring, 11,040 summer, 10,920 autumn and 10,800 winter hours.

The validation partition, by contrast, spans ten calendar months and contains no June and no July. This asymmetry is a consequence of requiring a full-year test period within a record of this length and is noted because it constrains any procedure calibrated on the validation partition: the summer months absent from it are also the months of highest irradiance and lowest day-to-day variability.

12. Intrinsic predictability

To establish the difficulty of the forecasting problem independently of any learned model, two non-parametric reference forecasts were evaluated on the test partition: **persistence**, which repeats the value observed 24 h earlier, and **climatology**, the mean of the corresponding (site, month, hour) cell computed on the training partition alone. Both were scored through the identical windowing and evaluation procedure applied to the models.

Table 12. Reference forecast accuracy on the test partition, pooled over sites.

Reference	Subset	RMSE (W/m ²)	MAE (W/m ²)	R ²
Persistence	All 24 hours	86.67	36.72	0.902
Climatology	All 24 hours	78.33	38.53	0.920

Reference	Subset	RMSE (W/m ²)	MAE (W/m ²)	R ²
Persistence	Daylight	121.56	72.15	0.811
Climatology	Daylight	109.86	75.72	0.846

Three properties of this table govern how the model results in this paper should be read.

First, night observations inflate every metric. The same climatological reference attains RMSE 78.3 W/m² and R² 0.920 when evaluated over all hours, but RMSE 109.9 W/m² and R² 0.846 over daylight hours. Including night observations reduces RMSE by 29% and raises R² by 0.074 without any contribution from the forecast. Because R² is normalised by the variance of the subset over which it is computed, and that variance is dominated by the day-night oscillation, an all-hours R² above 0.9 on this dataset is not evidence of forecast skill. Daylight statistics are reported throughout this work as the primary results.

Second, the reference to beat is climatology, not persistence. At a horizon of 24 h, persistence is aligned with the diurnal cycle and therefore inherits the deterministic geometric component at no cost; exceeding it is not in itself informative. Climatology is the stronger reference on RMSE and R².

Third, no single reference dominates. Climatology is the stronger of the two on RMSE and R², while persistence is stronger on MAE, reflecting the asymmetry of the error distribution. A forecast must therefore be shown to improve on 109.86 W/m² in RMSE, 0.846 in R² and 72.15 W/m² in MAE before it can be said to have improved on the naive references at all.

Site-level reference accuracy quantifies the regime difference of Section 7. The climatological reference attains a daylight RMSE of 97.2 W/m² at Antalya, 98.8 at Van, 107.1 at Konya and 108.2 at Ankara, but 133.8 at Rize, with R² falling from 0.88 to 0.71. Figure 8 shows this comparison. The pooled figure, in which four similar sites outvote one dissimilar one, understates the difficulty Rize presents.

Reference forecast floor: 24 h ahead, the model's test window, daylight hours

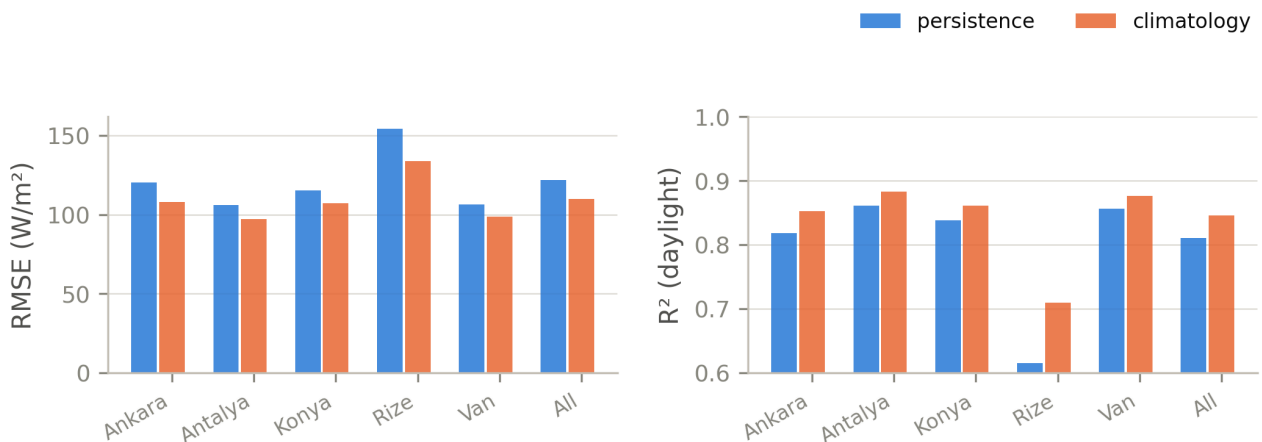


Figure 8. Accuracy of the naive reference forecasts on the test partition, by site, over daylight hours.

13. Summary

The dataset comprises 303,240 hourly observations distributed evenly over five Turkish sites and approximately seven years, with no missing values, one radiation variable and six meteorological predictors. Its principal characteristics, in the order in which they constrain the analysis, are:

1. Approximately half of all observations are night hours of exactly zero irradiance, and they must be excluded from evaluation for reported metrics to be interpretable.
2. Timestamps follow per-site local solar time rather than a common zone, so hourly quantities cannot be compared across sites.
3. Four sites form a single radiative regime and the fifth, Rize, forms a second one that is both less clear and substantially less predictable.
4. Irradiance and its predictability vary inversely over the year, with winter day-to-day variability 1.8–3.8 times that of summer.
5. Useful autoregressive information at the daily scale is essentially confined to the preceding day.
6. Most raw associations between meteorological variables and irradiance are confounded with solar geometry; three of six reverse sign once geometry is controlled for.
7. The naive reference forecasts attain a daylight RMSE of 109.86 W/m² and an R² of 0.846, which is the threshold any learned model must exceed to constitute a result.